

NIRANJAN PUNACHA B A

AI & MACHINE LEARNING ENGINEERING STUDENT

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PROFESSIONAL SUMMARY

Innovative AI & Machine Learning Engineering student specializing in Natural Language Processing (NLP), backend automation, and computer vision. Skilled in building scalable, AI-driven applications and intelligent automation solutions using Python, Flask, MongoDB, and LLM integrations. Strong foundation in Data Structures and Algorithm design, with a proven track record of developing real-world, full-stack AI products from concept to deployment. Actively seeking opportunities to leverage technical proficiency in a challenging engineering role.

TECHNICAL SKILLS

Programming Languages: Python, Java, C, C++

AI & Machine Learning: Deep Learning, TensorFlow, PyTorch, Gemini AI, LLM Integration

Data & Databases: NumPy, Pandas, MongoDB, MySQL, SQLite

Backend & Tools: Flask, REST APIs, Git, GitHub

Core Fundamentals: Data Structures & Algorithms, Object-Oriented Programming (OOP), Multithreading

Computer Vision: OpenCV

EDUCATION

Bachelor of Engineering (B.E.) in Artificial Intelligence and Machine Learning

Expected Graduation: 2027

Coorg Institute of Technology, Kodagu

TECHNICAL PROJECTS

Data-to-Database Automation using AI

Python, Flask, Gemini AI, MongoDB

- Developed a comprehensive system architecture for an AI-powered web application that utilizes NLP to convert voice and text input into structured JSON data.
- Integrated Gemini LLM to process natural language and implemented dynamic, schema-less document creation in MongoDB via a RESTful Flask backend.
- Automated complex data entry workflows, significantly improving operational efficiency by transforming conversational input directly into structured database records.

AI Receptionist System

Python, LLM Integration, API Automation

- Built an AI-powered virtual receptionist capable of autonomously parsing and handling complex user queries.
- Integrated SMS automation APIs to trigger real-time alerts, notifications, and appointment confirmations.
- Streamlined communication workflows and enhanced user experience through intelligent, logic-based task automation.

Sign Language to Text Communication System

Python, OpenCV, Machine Learning

- Developed a real-time computer vision system utilizing Python and OpenCV for accurate sign language recognition.
- Trained and implemented machine learning algorithms to map specific hand gestures and convert them into readable text output.
- Engineered a visual processing pipeline aimed at improving communication accessibility for individuals with hearing impairments.

Automated Trading Bot System

Python, Data Analysis, Algorithmic Backtesting

- Engineered an automated trading bot project using Python to implement a moving average crossover strategy.
- Designed and executed technical indicators to evaluate market trends and trigger automated trading signals.
- Documented project objectives and conducted rigorous performance evaluation of the trading bot through historical data backtesting.

CERTIFICATIONS

- PyTorch & TensorFlow Certification — Thanya Tech (Jan 2026)
- Software Engineering & Agile Development (May 2025)
- Computer Networks & Internet Security (Feb 2025)
- Multithreading in Java (Nov 2024)
- Programming in C & Basics of Python (Sep 2024)
- Infosys Springboard Certification (Passed with Distinction)

ADDITIONAL STRENGTHS

Algorithm Design & Problem Solving | Cross-functional Team Collaboration